OpenCode Student Setup Guide

Complete, zero-cost guide to autonomous AI coding in your terminal.

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github.com/Harman209/ecoloop

1. What is OpenCode?

OpenCode is an open-source terminal agent developed by Anomaly (GitHub: `anomalyco/opencode`). Unlike typical chatbots where you copy and paste code back and forth, OpenCode inspects your project directory, edits files directly, and runs your compiler or tests to automatically fix errors.

2. Quick Installation (60 Seconds)

Open your terminal on Linux, macOS, or Windows (WSL / PowerShell) and run:

```
# Install OpenCode globally via npm
npm install -g @opencode/cli

# Alternatively, install via curl (macOS / Linux)
curl -fsSL https://opencode.ai/install.sh | bash
```

3. Free AI Models: Zero Cost, No Credit Card Required

You do not need paid subscriptions to use autonomous AI agents. Here are the best free options available to students today:

Google Gemini API (Cloud, Free)

Sign in to **aistudio.google.com** with your Google account and generate a free API key. Generous free limits for Gemini 1.5 Flash and Gemini 2.0.

Groq Cloud (Cloud, Free)

Sign in to **console.groq.com**. Extremely fast token generation running open-weight models like Llama 3.3 70B for zero dollars.

GitHub Models (Cloud, Free)

Accessible directly from your GitHub student account. Test models like GPT-4o and Claude 3.5 Sonnet directly in VS Code or GitHub Codespaces.

Ollama (100% Offline, Private)

Download **ollama.com**. Runs entirely on your laptop hardware without an internet connection. Recommended model: `qwen2.5-coder:7b`.

4. Recommended Daily Workflow

1. **Navigate to your code:** In your terminal, go to any project folder (e.g. `cd ~/my-app`).
2. **Launch OpenCode:** Type `opencode` and hit enter.
3. **Choose your mode:** Use `plan` mode to survey files and review architectural options before any code changes. Switch to `build` mode to let the agent apply changes across your repository.
4. **Define clear rules:** Create a short file called `AGENTS.md` in your project root. Document your tech stack, design guidelines, and constraints so the agent never goes off track.